

Coleman Lai

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Education

BSc, Major in Computing Science

Sep 2023 – Present

Simon Fraser University, Burnaby, Canada | GPA: **4.00**

Dean's Honour Roll × 2, President's Honour Roll, SFU Open Scholarship × 3

Experience

Software Developer, Gen AI Co-op | IFS Copperleaf

Sep 2025 – Apr 2026

- Built an AI agent using Azure AI Foundry C# SDK capable of intelligently retrieving product documentation via [vector stores and search indexes with automatic source citations](#), dramatically speeding up knowledge discovery
- Deployed features through CI/CD workflows, accelerating feature rollout and maintaining high-quality releases

Developer | SFU Surge (Student Organization)

Jul 2025 – Present

- Developed full-stack, secure websites for cases including a [hackathon sites and portals for 1000+ participants](#)
- Worked with [product and visual design teams](#) through Figma and Notion to design and build applications

Website Administrator and Designer | iListen Canada

Jul – Oct 2024

- Upgraded company website on clean design principles, ensuring [alignment with company values](#) and increasing appeal
- Achieved 99%+ Lighthouse scores across SEO, performance, and accessibility through targeted site optimizations

Projects

Torgal | nwHacks | Electron, Python, PyTorch, faster-whisper | [🔗](#)

Jan 2026

- Built an [on-device presentation tool](#) that automatically advances slides by analysing speech, reducing pausing by 50%
- Architected a [semantic slide-matching engine](#) using sentence-transformer embeddings with adjacency biasing, keyword hotword boosts, and LRU-cached speech windows for responsive real-time transitions
- Engineered an [Electron–Python IPC pipeline](#) streaming speech with separate CPU/CUDA Windows build packaging

Celestial Maze | SFU Surge | Svelte, TypeScript, Firebase | [🔗](#)

Aug – Oct 2025

- Created a [hackathon-promoting dungeon crawling game](#) with procedurally generated mazes and enemy pathfinding
- Architected a [constraint-based maze generator](#) using graph theory to validate room connectivity and path reachability
- Optimized maze generation and rendering pipeline with object pooling, delta-time game loops, and spatial grid collision detection achieving <200ms load times and consistent 60 FPS performance across all devices

grid-detection | Python, PyTorch, NumPy, Pillow | [🔗](#)

Jul 2025

- Developed a [machine learning-based computer vision pipeline](#) to infer grid sizes from origami box-pleated crease patterns, allowing for detection of 23 of the most common grid sizes in an accurate and rapid manner
- Used a previously [self-developed Python library](#) for crease pattern manipulation to aid the preprocessing pipeline
- Generated synthetic training data through a randomized algorithm simulating realistic crease pattern structures

Hermes | Hugging Face MCP Hackathon | React, TypeScript, Python, Node, Docker, MCP, PostgreSQL | [🔗](#)

Jun 2025

- Programmed a [Model Context Protocol](#)-enabled tool that [automates asynchronous API calls](#) with configurable functions, simplifying API monitoring, data logging, and performance analysis through secure authorization steps
- Structured [adaptive API handling](#) in Python, including parsing of header tokens and complex user parameters
- Linked functions to SQL database operations, ensuring properly handled edge cases and data security

Skills

Languages: Python, JavaScript, TypeScript, HTML/CSS, C, C++, SQL, C#, R

Programs/Tools: Visual Studio, VSCode, Git, Azure SDK, OpenAI API, MCP, Docker, pytest, Websocket, Unity

Libraries/Frameworks: PostgreSQL, MongoDB, Node.js, React, Svelte, Angular, TailwindCSS, ExpressJS, Azure Portal

Other: Agile, Scrum, DevOps, DSA, CI/CD, Software Development Lifecycle (SDLC), Socket Programming